SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Experiment - 3

Aim: To estimate intrinsic and extrinsic camera parameters using checkerboard images.

Software Tools Required: VS Code, Python 3.12

Theory:

Camera calibration is a fundamental process in computer vision used to determine the internal characteristics (intrinsic parameters) and the spatial position and orientation (extrinsic parameters) of a camera. Accurate calibration is essential for interpreting the geometric structure of a scene from 2D images, correcting distortions, and enabling accurate measurements, 3D reconstruction, and navigation tasks.

A common and effective approach for camera calibration involves using multiple images of a known geometric pattern—typically a **checkerboard**—captured from different viewpoints. This provides a relationship between the known 3D geometry of the pattern and its corresponding 2D projections on the image plane.

1. Intrinsic Parameters

Intrinsic parameters describe the internal geometry of the camera, including:

- **Focal length (*fx, fy*):** Represents the scaling factors in the image axes and is typically measured in pixels. It relates to the physical focal length of the lens and the pixel size of the sensor.
- **Principal point (*cx, cy*):** Denotes the intersection of the optical axis with the image plane. In ideal cases, it is the center of the image.
- **Skew coefficient (s):** Describes the angle between the x and y pixel axes. For most modern cameras, the skew is zero due to rectangular pixels.
- **Distortion coefficients (*k1, k2, k3, p1, p2*):** Represent lens distortions. Radial distortion (*k1–k3*) causes straight lines to curve, especially near image edges, while tangential distortion (*p1, p2*) arises from misalignment between the lens and the sensor.

The intrinsic camera matrix K is

$$K = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

2. Extrinsic Parameters

Extrinsic parameters describe the camera's position and orientation in the world coordinate system. They consist of:

- **Rotation matrix (R):** Describes the camera's orientation.
  Translation vector (t): Describes the camera's position.

These parameters transform 3D world coordinates into the camera coordinate system.

3. Calibration Using a Checkerboard

Checkerboard patterns are ideal for calibration because their corners are easy to detect accurately and have a known, regular geometry.

Calibration Steps:

1. **Image Capture:** Acquire multiple images of the checkerboard from different angles, distances, and orientations to cover a wide field of view.
2. **Corner Detection:** Use algorithms like cv2.findChessboardCorners() to detect inner corners of the checkerboard in each image.
3. **Object-Image Point Mapping:** Assign real-world coordinates (e.g., (0,0,0), (1,0,0), ...) to each detected 2D corner point, assuming a fixed square size and Z=0 (flat checkerboard).
4. **Estimation:** Use algorithms like **Zhang's method** or **Direct Linear Transformation (DLT)** to solve for the camera parameters.
5. **Optimization:** Apply **non-linear optimization (e.g., Levenberg-Marquardt)** to minimize the reprojection error (the difference between observed and projected image points).
6. **Reprojection Error Analysis:** A low mean reprojection error (e.g., < 0.5 pixels) indicates accurate calibration.

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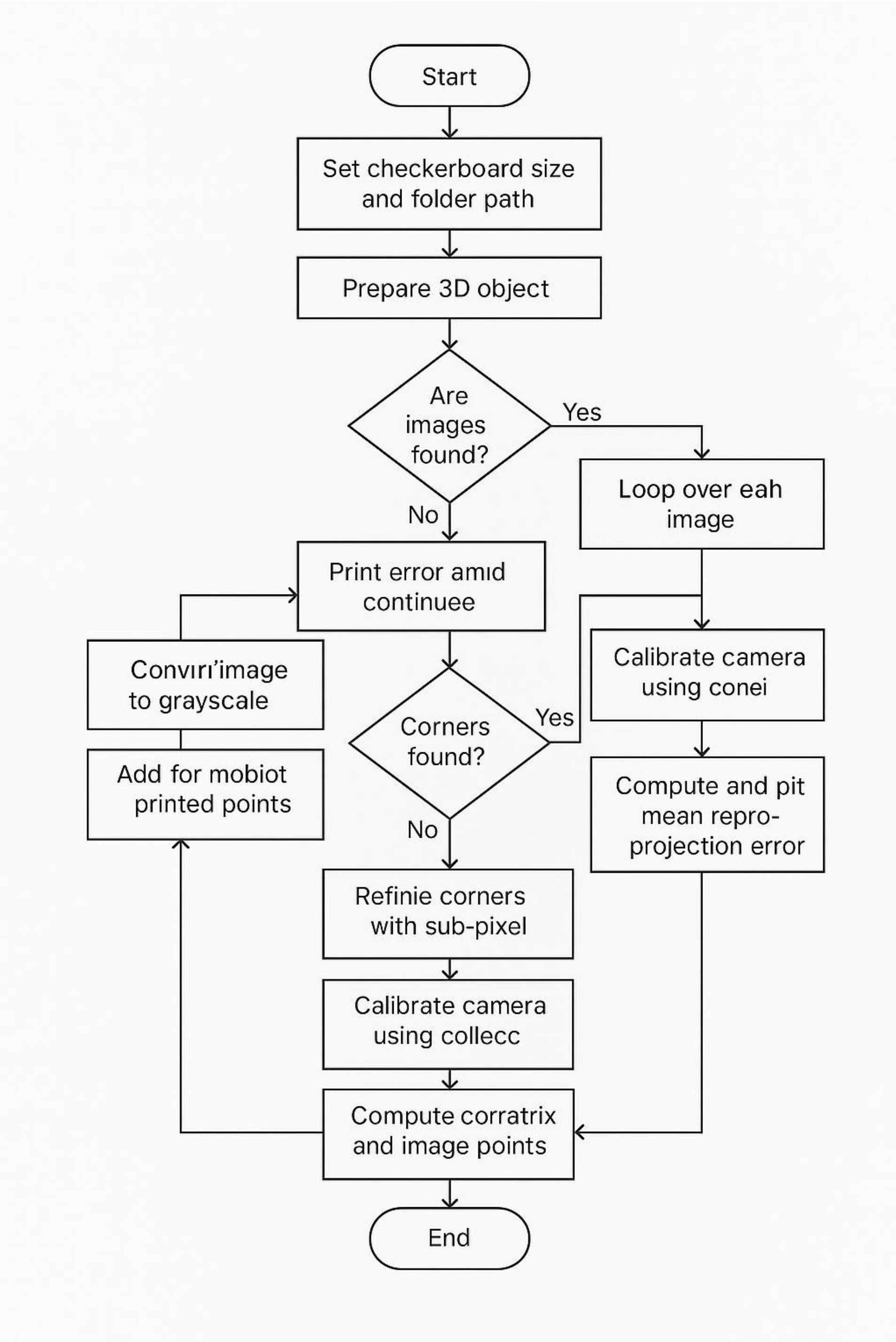
Flowchart


Start
Set checkerboard size and folder path
Prepare 3D object
Are images found?
Yes
No
Loop over eah image
Print error amıd continuee
Convırı'image to grayscale
Calibrate camera using conei
Corners found?
Yes
Add for mobiot printed points
Compute and pit mean repro-projection error
No
Refinie corners with sub-pixel
Calibrate camera using collecc
Compute corratrix and image points
End

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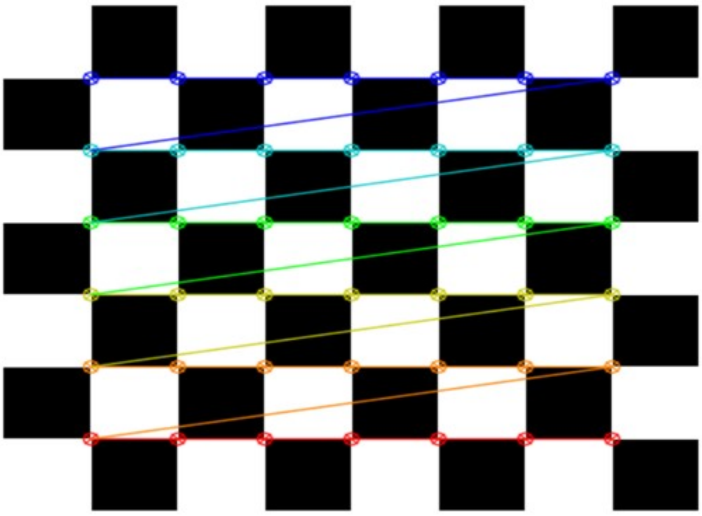
Constituent of Symbiosis International (Deemed University), Pune **Code:**

Department of Electronics and Telecommunication Engineering

```python
import cv2
import numpy as np
import glob
import os

# === CONFIGURATION ===
CHECKERBOARD = (7, 6)  # (columns, rows) of inner corners
image_folder = 'checkerboard_images'  # Folder containing checkerboard images

# === PREPARE OBJECT POINTS ===
objp = np.zeros((CHECKERBOARD[0] * CHECKERBOARD[1], 3), np.float32)
objp[:, :2] = np.mgrid[0:CHECKERBOARD[0], 0:CHECKERBOARD[1]].T.reshape(-1, 2)

objpoints = []  # 3D points in real world
imgpoints = []  # 2D points in image plane

# === READ IMAGES ===
images = glob.glob(os.path.join(image_folder, '*.jpg'))

if not images:
    print("❌ No images found in the folder. Please check the path or add images.")
    exit()

for fname in images:
    img = cv2.imread(fname)
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    ret, corners = cv2.findChessboardCorners(gray, CHECKERBOARD, None)

    if ret:
        objpoints.append(objp)
        corners2 = cv2.cornerSubPix(gray, corners, (11,11), (-1,-1),
                                    criteria=(cv2.TERM_CRITERIA_EPS + cv2.TERM_CRITERIA_MAX_ITER, 30, 0.001))
        imgpoints.append(corners2)

        cv2.drawChessboardCorners(img, CHECKERBOARD, corners2, ret)
```

```python
        cv2.imshow('Detected Corners', img)
        cv2.waitKey(200)

cv2.destroyAllWindows()

# === CAMERA CALIBRATION ===
ret, camera_matrix, dist_coeffs, rvecs, tvecs = cv2.calibrateCamera(
    objpoints, imgpoints, gray.shape[::-1], None, None)

# === OUTPUT ===
print("\n === CAMERA CALIBRATION RESULTS ===")
print("\n Intrinsic Matrix (Camera Matrix):\n", camera_matrix)
print("\n Distortion Coefficients:\n", dist_coeffs)

for i, (rvec, tvec) in enumerate(zip(rvecs, tvecs)):
    print(f"\n Image {i+1} Extrinsic Parameters:")
    print("  Rotation Vector:\n", rvec)
    print("  Translation Vector:\n", tvec)

# === REPROJECTION ERROR ===
mean_error = 0
for i in range(len(objpoints)):
    imgpoints2, _ = cv2.projectPoints(objpoints[i], rvecs[i], tvecs[i], camera_matrix, dist_coeffs)
    error = cv2.norm(imgpoints[i], imgpoints2, cv2.NORM_L2)/len(imgpoints2)
    mean_error += error
print(f"\n Mean Reprojection Error: {mean_error/len(objpoints):.4f}")
```

Output:



```
=== CAMERA CALIBRATION RESULTS ===

Intrinsic Matrix (Camera Matrix):
[[2.56845336e+06 0.00000000e+00 3.99500075e+02]
[0.00000000e+00 2.56604928e+06 2.99499974e+02]
[0.00000000e+00 0.00000000e+00 1.00000000e+00]]

Distortion Coefficients:
[[ 9.69105947e-05 -1.64131616e-12  2.93232268e-03 -3.15102435e-03
 -1.56764952e-20]]

Image 1 Extrinsic Parameters:
 Rotation Vector:
[[ 6.81125812e-02]
[-6.44068303e-05]
[-3.14085508e+00]]
 Translation Vector:
[[3.00550596e+00]
[2.50835684e+00]
[4.27674421e+04]]

Mean Reprojection Error: 0.0005
```

Conclusion:

This experiment uses checkerboard images, an effective method for estimating both intrinsic and extrinsic parameters during camera calibration. It helps determine the camera's internal properties and spatial position, enabling accurate 3D measurements and supporting essential computer vision tasks such as pose estimation and augmented reality.